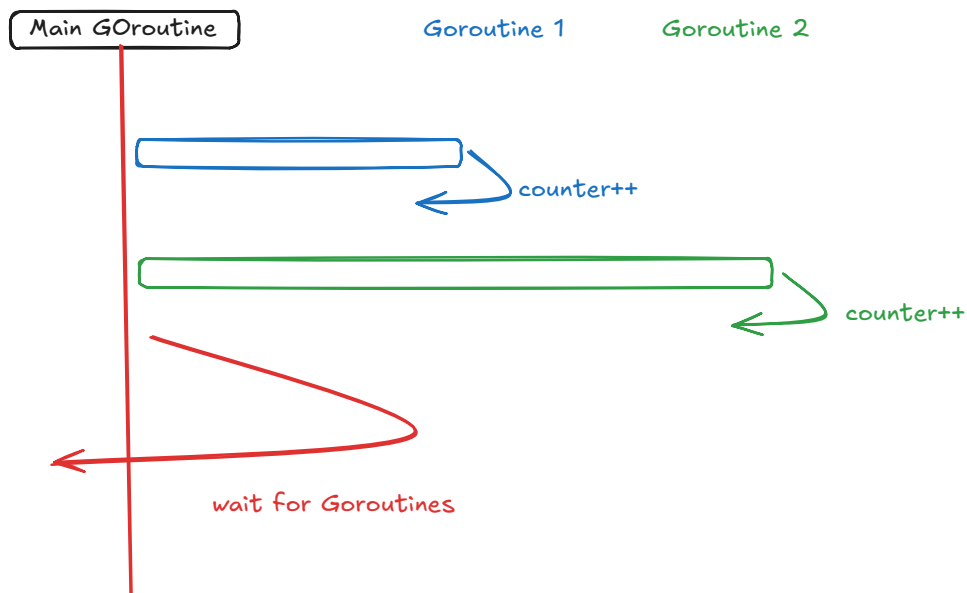


Race condition

A race condition occurs when the outcome of a program depends on the sequence or timing of uncontrollable events. If multiple threads or goroutines read and write shared data without proper synchronization, the final state of the data can vary depending on the timing of these operations. This can lead to inconsistent results, data corruption, and other unexpected behaviors.



Race conditions can manifest in various ways, often making them difficult to diagnose.

- Intermittent Bugs: bugs that occur sporadically and are hard to reproduce consistently.
- Unexpected Behavior: program behavior that deviates from the expected outcome, especially under concurrent execution.
- Crashes and Panics: sudden crashes or panics that occur without a clear cause, often due to corrupted state.